

Point-of-Care Edge-AI Haematology Screening Report

Automated Smear Triage & Morphological Analysis — MIT-WPU Research Group

Patient ID / Sample:	PAT-2026-0841	Acquisition Date:	2026-09-16
Imaging Modality:	Compound Microscope (100x Oil)	Edge Processor:	Raspberry Pi 4 Model B (INT8)
Screening Status:	Normal Morphology	Sensor:	RPi HQ Camera (12.3 MP)

Quantitative Cytological Counts & Differential Distribution:

Cell Type	Detected Count (per FOV)	Differential (%)	Reference Range (Smear)
Red Blood Cells (RBC)	52	-	Normocytic, Normochromic
White Blood Cells (WBC)	3	100.0%	4.0 - 11.0 x 10 ³ / uL
Platelets (Thrombocytes)	7	-	Adequate (7-15 per 100x FOV)

DIAGNOSTIC STATUS: Standard cellular morphology. No acute sickling detected.

Explainability Metric: Average prediction entropy = 0.14. 0 ambiguous cell region(s) flagged for manual pathologist review under active learning protocol.